Across CES

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Abstract

This paper studies demand systems through their associated budget share functions and provides a representation for demand systems that generalize Constant Elasticity of Substitution (CES). The key observation is that, in CES demand systems, each own-price share coincides pointwise with the cumulative or complementary cumulative distribution function of a log-logistic distribution. Building on this observation, the paper shows that any demand system that nests CES induces share functions that can be represented as generalizations of log-logistic (C)CDFs. Moreover, if generalized share functions are monotone in own price and admit limits at zero and infinity, they themselves coincide with (complementary) distribution functions. The result is purely representational and does not rely on utility maximization or integrability. It provides a constructive framework for designing and interpreting demand systems beyond CES, clarifies the structure of existing generalizations, and offers a direct representational link between continuous-good demand systems and distribution-based representations commonly used in choice theory.

1 Introduction

Constant Elasticity of Substitution (CES) demand systems are ubiquitous in economic modeling. Their appeal lies in tractability: CES delivers closed-form demands, a simple price index, and transparent comparative statics. As a result, CES demand appears as a building block across fields ranging from international trade and industrial organization to macroeconomics and public finance.

At the same time, the structural restrictions imposed by CES are well known. CES enforces uniform substitution patterns across goods, rules out local complementarity, and fixes curvature in ways that are often at odds with empirical evidence or with the needs of applied modeling. A large literature therefore proposes demand systems that move *beyond CES*, including nested CES structures, variable-elasticity aggregators, and more flexible functional forms for indirect utility or expenditure functions.

This paper takes a different perspective. Rather than modifying primitives such as utility or price aggregators, it studies *budget shares* directly. Any demand system $\mathbf{x}(\mathbf{p}, Y)$ can be written in the form

$$x_j(\mathbf{p}, Y) = \frac{Y}{p_j} s_j(\mathbf{p}, Y),$$

where the share functions $\mathbf{s}(\cdot)$ satisfy positivity and adding-up. From this viewpoint, CES is characterized by the fact that each share s_j admits a simple closed-form expression in own price, conditional on the rest of the price vector.

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The key observation of the paper is that, in CES demand systems, the own-price share function admits an exact representation in the language of probability theory: holding $(\mathbf{p}_{-j}, Y)$ fixed, $s_j(p_j)$ coincides pointwise with the (complementary) cumulative distribution function of a log-logistic distribution.[1] This fact is elementary but powerful. It suggests that CES demand systems can be understood as arising from a particular choice of distributional shape for own-price shares.

The main contribution of the paper is to show that this perspective extends well beyond CES. Any demand system that generalizes a CES demand system—in the sense that CES is recovered for some parameter values or limits—induces share functions that can be represented as generalizations of the log-logistic (C)CDF. Moreover, if share functions satisfy a natural monotonicity condition and admit limits at zero and infinity in own price, then they are themselves (complementary) CDFs. In this sense, demand systems that generalize CES can be characterized by families of distribution functions.

This result should be understood as a representation theorem. It does not introduce new primitives, nor does it restrict attention to utility-maximizing demand systems. Rather, it provides a systematic way of parameterizing and constructing demand systems through the choice of admissible CDF-like objects. Seen this way, CES appears as one particular parametrization within a broader design space.

The representation is useful for several reasons. First, it provides a constructive method for generating new demand systems: by choosing alternative functional forms for the underlying distribution or for its generalization, one obtains demand systems that nest CES while allowing for richer patterns of substitution. Second, the approach clarifies the structure of existing generalizations of CES. In particular, demand systems such as the Homothetic Single Aggregator (HSA) and Homothetic Directly Aggregable (HDIA) classes can be rewritten as generalized log-logistic share systems. Third, the representation offers a natural bridge between continuous-good demand systems and the probabilistic language commonly used in discrete-choice models, where market shares are interpreted as choice probabilities derived from distribution functions.

Importantly, the analysis is deliberately agnostic about integrability and rationalization. While homogeneity and adding-up are discussed explicitly, the paper does not impose utility maximization as a maintained assumption. Instead, it characterizes the shape of share systems themselves. Conditions under which CDF-based share systems can be rationalized by an underlying utility function are deferred to future work and briefly discussed in the Appendix.

The paper proceeds as follows. Section 2 introduces the general framework for demand systems and establishes the main representation result. Section 3 applies the characterization to existing classes of demand systems and illustrates how new demand systems can be constructed within the framework. An example allowing for localized complementarity is also provided. The Appendix discusses additional technical issues and extensions, including equilibrium existence, rationalizability, and global domain properties.

[1] This interpretation is purely functional and does not rely on stochastic choice or random utility; it is used solely as a representational device.

2 Demand Systems

Consider an n-dimensional consumption space. Let $\mathbf{x}, \mathbf{p} \in \mathbb{R}^n_{++}$ denote consumption bundles and prices, and let $Y \in \mathbb{R}_{++}$ denote income. A demand system is a mapping

$$\mathbf{x} : \mathbb{R}^n_{++} \times \mathbb{R}_{++} \to \mathbb{R}^n_{++}.$$

2.1 Budget shares as primitives

Any demand system can be written in the form

$$x_j(\mathbf{p}, Y) = \frac{Y}{p_j} s_j(\mathbf{p}, Y), \tag{1}$$

where the functions $s_j(\cdot)$ satisfy

$$s_j(\mathbf{p}, Y) > 0, \qquad \sum_{j=1}^{n} s_j(\mathbf{p}, Y) = 1.$$

The functions $\mathbf{s}(\cdot)$ are referred to as *budget share functions*. Equation (1) is an identity and does not impose behavioral restrictions such as utility maximization or integrability. Throughout the paper, demand systems are therefore studied directly at the level of their associated share functions.

Given a collection of share functions $\mathbf{s}(\cdot)$, define the set

$$E \equiv \left\{ (\mathbf{p}, Y) \in \mathbb{R}^{n+1}_{++} \ : \ \sum_{j=1}^{n} s_j(\mathbf{p}, Y) = 1 \right\}.$$

For any $(\mathbf{p}, Y) \in E$, the associated demand system satisfies the budget constraint $Y = \sum_j p_j x_j(\mathbf{p}, Y)$. The set E represents the combinations of prices and income that the demand system can accommodate without violating adding-up. Whether E coincides with the full domain $\mathbb{R}^{n+1}_{++}$ or is a strict subset depends on the structure of the share functions.

2.2 CES demand systems and own-price structure

In a CES demand system, each share function takes a particularly simple form. Fix j and hold $(\mathbf{p}_{-j}, Y)$ constant. Then the CES share $s_j(p_j; \mathbf{p}_{-j}, Y)$ can be written as

$$s_j(p_j; \mathbf{p}_{-j}, Y) = \frac{1}{1 + p_j^{\sigma-1} \sum_{k \neq j} \left(\frac{\alpha_k}{\alpha_j} \right)^{\sigma} p_k^{1-\sigma}}, \tag{2}$$

where $\sigma > 0$ and $\alpha_j > 0$.

Viewed as a function of own price p_j, expression (2) has three properties. First, it maps $\mathbb{R}_{++}$ into $(0, 1)$. Second, it is monotone in p_j. Third, it admits well-defined limits as $p_j \to 0$ and $p_j \to \infty$. As a result, holding $(\mathbf{p}_{-j}, Y)$ fixed, the CES share coincides pointwise with either a cumulative distribution function or a complementary cumulative distribution function of a log-logistic distribution, depending

on whether $\sigma < 1$ or $\sigma > 1$.[2]

This observation is purely representational. No randomness or stochastic choice is introduced. The relevance of the distributional language is that it provides a convenient and structured way of describing the shape of own-price share functions.

2.3 Generalizing CES

The central object of interest in this paper is a class of demand systems that *generalize CES*. Formally, consider a family of demand systems $\{\tilde{\mathbf{x}}(\mathbf{p}, Y; \boldsymbol{\gamma}) : \boldsymbol{\gamma} \in \boldsymbol{\Gamma}\}$. The system is said to generalize a CES demand system $\mathbf{x}^{\text{CES}}(\mathbf{p}, Y)$ if there exists $\tilde{\boldsymbol{\gamma}} \in \overline{\boldsymbol{\Gamma}}$ such that

$$\tilde{\mathbf{x}}(\mathbf{p}, Y, \tilde{\boldsymbol{\gamma}}) = \mathbf{x}^{\text{CES}}(\mathbf{p}, Y) \quad \text{for all } (\mathbf{p}, Y) \in \mathbb{R}^{n+1}_{++}.$$

No further structure is imposed. In particular, the generalized system is not required to preserve homogeneity, separability, or any other property of CES away from $\tilde{\boldsymbol{\gamma}}$.

For clarity, the notion of generalization used throughout is purely functional. A family of demand systems $\{\tilde{\mathbf{x}}(\cdot; \boldsymbol{\gamma})\}$ is said to generalize a CES demand system if CES is recovered for some parameter value or limit $\tilde{\boldsymbol{\gamma}} \in \overline{\boldsymbol{\Gamma}}$. Equivalently, the associated share functions $\{\tilde{\mathbf{s}}(\cdot; \boldsymbol{\gamma})\}$ generalize CES shares if $\tilde{s}_j(\cdot; \tilde{\boldsymbol{\gamma}}) = s_j^{\text{CES}}(\cdot)$ pointwise for each j. No additional structure is imposed on the family away from $\tilde{\boldsymbol{\gamma}}$. The following theorem characterizes the structure of such share systems.

Theorem 1. *Suppose $\tilde{\mathbf{x}}(\mathbf{p}, Y, \boldsymbol{\gamma})$ generalizes a CES demand system. Then:*

(i) For each good j, the CES share $s_j^{\text{CES}}(p_j; \mathbf{p}_{-j}, Y)$ coincides with a (complementary) cumulative distribution function of a log-logistic distribution in own price p_j.

(ii) For each j, the generalized share $\tilde{s}_j(p_j; \mathbf{p}_{-j}, Y, \boldsymbol{\gamma})$ coincides with a generalization of this log-logistic function.

(iii) If $\tilde{s}_j(\cdot)$ is monotone in p_j and admits limits at zero and infinity, then $\tilde{s}_j(\cdot)$ itself coincides with a (complementary) cumulative distribution function.

Theorem 1 provides a representation result. Any demand system that nests CES induces own-price share functions that can be described as generalized distribution functions. Conversely, specifying admissible CDF-like objects generates demand systems that nest CES by construction. The theorem does not rely on integrability or on utility maximization and is therefore applicable to a broad class of demand systems.

2.4 Constructing generalized share systems

The representation in Theorem 1 is constructive. For example, a direct generalization of (2) is given by

$$\tilde{s}_j(p_j; \mathbf{p}_{-j}, Y) = \frac{1}{1 + \frac{h(p_j)}{p_j} \sum_{k \neq j} \frac{p_k}{h(p_k)}}, \tag{3}$$

[2] In the CES case, the distinction between a cumulative and a complementary cumulative distribution function coincides with the familiar split between the $\sigma < 1$ (complements) and $\sigma > 1$ (substitutes) regimes. The distributional language used here (CDF vs. CCDF) is purely representational and is not intended as a general classification of gross substitutes versus gross complements.

where $h : \mathbb{R}_{++} \to \mathbb{R}_{++}$ is strictly increasing. If $h(p) = p^\sigma$, expression (3) reduces to the CES share with symmetric taste parameters. More generally, the choice of $h(\cdot)$ governs the curvature and substitution patterns of the demand system.

The next section applies this representation to existing classes of demand systems and illustrates how it can be used to construct new ones.

3 Examples and Applications

This section illustrates how the representation result in Theorem 1 can be used both to reinterpret existing demand systems and to construct new ones. The emphasis is not on deriving demand from primitives, but on understanding the admissible structure of demand systems directly at the level of budget shares.

3.1 A construction recipe

Theorem 1 provides a simple design principle. To construct a demand system that nests CES, it suffices to specify, for each good j, an own-price share function that generalizes the CES log-logistic shape and satisfies positivity and adding-up.

A convenient class of examples is given by

$$\tilde{s}_j(p_j; \mathbf{p}_{-j}, Y) = \frac{1}{1 + \frac{h(p_j)}{p_j} \sum_{k \neq j} \frac{p_k}{h(p_k)}},$$

where $h : \mathbb{R}_{++} \to \mathbb{R}_{++}$ is strictly increasing. When $h(p) = p^\sigma$, the expression reduces to CES with symmetric taste parameters. More generally, the curvature of $h(\cdot)$ governs the sensitivity of shares to own prices and therefore the substitution patterns of the demand system.

This construction illustrates how the representation theorem converts the problem of specifying a demand system into the choice of an admissible CDF-like object. Different choices of $h(\cdot)$ generate different demand systems while preserving the CES benchmark as a special case.

3.2 Existing generalizations of CES

Several demand systems proposed in the literature can be directly rewritten in the form characterized by Theorem 1. This subsection illustrates the point using the Homothetic Single Aggregator (HSA) and Homothetic Directly Integrable Aggregator (HDIA) demand systems studied by **?**.

HSA. In HSA demand systems, zero homogeneity of demand is achieved through a linearly homogeneous price aggregator $A(\mathbf{p})$, so that

$$s_j(\mathbf{p}, Y) = s_j\left(\frac{p_j}{A(\mathbf{p})}\right).$$

As in CES, HSA demand systems cannot combine gross substitutes and gross complements, the admissible domain E^{HSA} is compact, and each share admits limits at zero and one over its own price within that domain. Rewriting the HSA shares using the adding-up condition shows that each $s_j(\cdot)$ can be expressed as a generalized log-logistic (C)CDF, consistent with the representation in Theorem 1.

HDIA. HDIA demand systems allow shares to depend not only on relative prices but also on levels of consumption. Using the original formulation in **?**, the shares can be written as

$$s_j(\mathbf{p}, Y) = \frac{1}{1 + \frac{1}{p_j x_j\left(\frac{p_j}{B(\mathbf{p})}\right)} \sum_{k \neq j} p_k x_k \left(\frac{p_k}{B(\mathbf{p})}\right)}.$$

Written in this form, the HDIA shares are seen to coincide with generalized log-logistic (C)CDFs. Thus, both HSA and HDIA demand systems fit naturally within the representation framework developed in this paper.

3.3 Localized complementarity

CES and the generalizations discussed so far impose a global structure in which all goods are either mutual gross substitutes or mutual gross complements. The representation framework also allows for more localized departures from this structure.

As an illustration, consider a demand system in which all goods are gross substitutes except for a single pair (j, l), which are gross complements with each other. Such a system can be constructed by modifying only the share functions for goods j and l, while leaving all other shares unchanged:

$$\begin{aligned}
s_l(p_l; \mathbf{p}_{-l}, Y) &= \frac{1}{1 + p_l^{\sigma-1} \sum_{k \neq l} \left(\frac{\alpha_k}{\alpha_l}\right)^{\sigma} p_k^{1-\sigma} + \gamma p_j^{\mu}}, \\
s_j(p_j; \mathbf{p}_{-j}, Y) &= \frac{1}{1 + p_j^{\sigma-1} \sum_{k \neq j} \left(\frac{\alpha_k}{\alpha_j}\right)^{\sigma} p_k^{1-\sigma} + f}, \\
s_i(p_i; \mathbf{p}_{-i}, Y) &= \frac{1}{1 + p_i^{\sigma-1} \sum_{k \neq i} \left(\frac{\alpha_k}{\alpha_i}\right)^{\sigma} p_k^{1-\sigma}}, \qquad i \neq j, l,
\end{aligned}$$

where $\mu > \sigma > 1$, $\gamma > 0$, and the adjustment term f is chosen to ensure adding-up.

This example demonstrates how the CDF-based representation can be used to introduce localized complementarity while preserving the CES structure elsewhere. A full analysis of existence, uniqueness, and smoothness of the implicit adjustment term is omitted and left for future work.

4 Conclusion

This paper studies demand systems through the lens of budget share functions and provides a representation for demand systems that generalize Constant Elasticity of Substitution (CES). The central observation is that, in CES demand systems, each own-price share coincides pointwise with the cumulative or complementary cumulative distribution function of a log-logistic distribution. Building on this fact, the paper shows that any demand system that nests CES induces share functions that can be represented as generalizations of log-logistic (C)CDFs. Under mild regularity conditions—namely own-price monotonicity and the existence of limits at zero and infinity—these generalized shares themselves coincide with (C)CDFs.

The contribution is deliberately representational. Rather than proposing new primitives or imposing utility maximization, the paper characterizes the functional structure of demand systems directly at the

level of budget shares. Seen from this perspective, CES appears as one particular parametrization within a broader class of admissible share systems. Existing generalizations of CES, such as the HSA and HDIA demand systems, fit naturally within this framework, while the representation also allows for the construction of new demand systems, including ones with localized departures from global substitution patterns.

The results highlight a close connection between continuous-good demand systems and distribution-based representations familiar from choice theory. In the present paper, this connection is purely functional and does not rely on stochastic choice. Nonetheless, it provides a useful language for describing curvature, price sensitivity, and substitution behavior in a unified way.

Several directions for future research follow naturally. One is to characterize conditions under which CDF-based share systems are integrable and admit a utility representation. Another is to study equilibrium existence and uniqueness in models that employ generalized share systems, particularly when substitution patterns vary locally. More broadly, the representation developed here suggests that distribution-based parameterizations may offer a flexible and transparent toolkit for demand modeling beyond CES.

Appendix

A. Equilibrium domain and adding-up

Throughout the paper, demand systems are studied through their associated budget share functions. Given a collection of shares $\mathbf{s}(\cdot)$, define the set

$$E \equiv \left\{ (\mathbf{p}, Y) \in \mathbb{R}^{n+1}_{++} \ : \ \sum_{j=1}^{n} s_j(\mathbf{p}, Y) = 1 \right\}.$$

For any $(\mathbf{p}, Y) \in E$, the induced demand system satisfies the budget constraint exactly.

In some classes of demand systems, including CES, the set E coincides with the full domain $\mathbb{R}^{n+1}_{++}$. In other cases, such as demand systems built around a single price aggregator, the admissible domain may be a strict subset of $\mathbb{R}^{n+1}_{++}$. The representation results in the main text are agnostic about the size of E. When constructing generalized share systems, ensuring that E is nonempty or coincides with the full domain requires additional restrictions on the functional form of the shares.

B. Integrability and rationalization

The analysis in the paper does not impose utility maximization or integrability as a maintained assumption. All results are stated directly in terms of demand systems and their associated budget share functions.

It is well known that not every collection of positive, adding-up shares can be rationalized by a utility function. Additional restrictions—such as Slutsky symmetry and negativity—are required for integrability. Determining necessary and sufficient conditions under which CDF-based share systems admit a utility representation is an important question, but one that lies outside the scope of the present paper.

That said, many commonly used demand systems that satisfy integrability, including CES and several of its generalizations, fit naturally within the CDF-based representation developed here. Establishing integrability conditions for broader classes of generalized distribution-based shares is left for future work.

C. Interpretation as survival functions

The representation of own-price share functions as cumulative or complementary distribution functions admits a useful interpretation. Holding $(\mathbf{p}_{-j}, Y)$ fixed, the share $s_j(p_j)$ can be viewed as a *survival function* in own price: it describes how the expenditure share allocated to good j declines as its price increases.

Under this interpretation, the curvature of the share function corresponds to the shape of an associated hazard rate. Different choices of generalized CDFs therefore correspond to different forms of price sensitivity and substitution behavior. This perspective is closely related to the interpretation of market shares as choice probabilities in discrete-choice models, though no stochastic choice or random utility is assumed here. The survival-function language is used solely as a representational device.

D. Localized modifications of substitution patterns

Section 3 presents examples in which substitution patterns are modified for specific pairs of goods while leaving the remaining structure unchanged. Such constructions typically require implicit adjustments to ensure adding-up and positivity of shares.

A complete analysis of existence, uniqueness, and smoothness of these implicit adjustments is omitted from the main text. Providing sufficient conditions under which such localized modifications are well defined for all $(\mathbf{p}, Y) \in \mathbb{R}^{n+1}_{++}$ is a natural extension of the present framework.